MO YUNXIANG

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EDUCATION

The Hong Kong University of Science and Technology Sep 2024 – Present
BSc, Triple Major in Computer Science, Mathematics & Electronic Engineering; Extended Major in Artificial Intelligence
GPA: 4.1 / 4.3
Honors:

- **University's Scholarship Scheme for Continuing Undergraduate Students** (Top 1%), 2024
- **S.S. Chern Class** – *Honor for top academic performance across all mathematics coursework.*
- **Dean's List Honor** (GPA above 3.7), 2024, 2025

Relevant Coursework: Algorithms, Statistical Inference, Machine Learning, Linear Algebra, Probability Theory *(A+ in all CS/Math courses)*

PUBLICATIONS

First-author paper, ACL 2026 [OpenReview]
"DixitWorld: Evaluating Multimodal Abductive Reasoning in Vision-Language Models with Multi-Agent Dixit Gameplay"
Yunxiang Mo, Tianshi Zheng, Qing Zong, Jiayu Liu, Baixuan Xu, Yauwai Yim, Chunkit Chan, Jiaxin Bai, Yangqiu Song. Advisor: Prof. Yangqiu Song.

- Proposed DixitArena, a multi-agent environment that tests hypothesis generation (storyteller) and hypothesis selection (listener) under imperfect information, and DixitBench, a static QA benchmark (252 items over three difficulty tiers) that isolates the listener task
- Revealed a storyteller–listener asymmetry across six VLMs: over 78% of storyteller rounds score zero while the best listener reaches 75.6% accuracy; small open-source models are more creative storytellers, large proprietary models stronger listeners
- Validated DixitBench as a lightweight proxy for DixitArena ($r = 0.947$); built a reproducible evaluation pipeline over open-source (local) and proprietary (API) VLMs with controlled distractor difficulty, and a 72B-parameter scaling ablation plus calibration/sensitivity analyses show the gap is structural

First-author paper, under review [arXiv]
"AgentIdeaBench: Benchmarking Scientific Ideation in the Agent Era"
Yunxiang Mo, Tianshi Zheng, Yisen Gao, Rui Wang, Newt Nguyen Kim Hue Nam, Kelvin Kiu Wai Tam, Jiaxin Bai, Yangqiu Song, Ginny Wong, Simon See.

- Built a multidisciplinary benchmark that evaluates scientific ideation under matched static-observation and active-exploration settings, scoring 33 LLMs across 40 subfields in five disciplines with literature-verified critics that judge originality against retrieved prior art
- Showed that active exploration exposes far more capability headroom: performance scales about twice as fast as under static observation, the gain is capability-gated toward the strongest models, and it comes from better grounding (feasibility, clarity, specificity) rather than originality
- Explored Scientific World Modeling, a generation-time loop that refines hypotheses through structured thought experiments; it helps mid-capability models but adds little for frontier models

Co-first-author paper, under review
"Robust Information-Gain Control for Active Agentic Reasoning under Approximate Beliefs"
Fang Wu*, Yunxiang Mo*, Aaron Tu, ..., Yangqiu Song, Jure Leskovec (16 authors). **Equal contribution.*

- Formalized Belief Trap Regions and showed that information-gain maximization locks LLM agents into self-reinforcing loops when their internal beliefs are only approximate
- Proposed Distributionally Robust Information Gain (DR-IG), a training-free decision-time objective that maximizes worst-case information gain around the current belief, with a verbalized belief proxy extracted from confidence-annotated outputs
- Across five LLM families, DR-IG cuts trap incidence to 0% at every tested shortlist size and recovers solve rates where IG-max solves nothing; in a real browser DOM it recovers a significant share of the 26.5% solve rate lost to belief approximation

Co-author paper, ICLR 2026 (Oral) [arXiv]
"ScaleCUA: Scaling Open-Source Computer Use Agents with Cross-Platform Data"
Zhaoyang Liu, Jingjing Xie, ..., Yunxiang Mo, ..., Wenhai Wang. *Contribution: data pipeline and cross-platform workflow components in the open-source codebase.*

RESEARCH & INTERNSHIP EXPERIENCE

University of Illinois Urbana-Champaign (UIUC) Oct 2026 – Present
Research Assistant Working with Prof. Jiaxuan You (U Lab)

KnowComp Group, HKUST Apr 2025 – Present
Undergraduate Researcher Advised by Prof. Yangqiu Song; mentored by Dr. Tianshi Zheng

- Led two first-author projects on reasoning evaluation, DixitWorld (ACL 2026) and AgentIdeaBench (under review), from problem formulation through environment and benchmark design to large-scale evaluation of open-source and proprietary models

Stanford HAI, Stanford University Apr 2026 – Present
Research Intern Mentored by Dr. Fang Wu in the groups of Prof. Yejin Choi and Prof. Jure Leskovec

- Robust, belief-aware decision-making for LLM agents in multi-turn reasoning (co-first-author DR-IG paper above)
- **RL for diffusion language models** (Jun – Jul 2026): led RL fine-tuning of LLaDA-8B-Instruct with 4-bit LoRA and DeepSpeed ZeRO-2 on 8×A100; implemented d-ReLo's block-level credit assignment along denoising trajectories with a learned value function, building on d1's diffu-GRPO code
- Designed controlled multi-seed comparisons against GRPO on Countdown and Sudoku across credit-assignment granularities; diagnosed value-function learning through learning-rate, warm-up and reward-form ablations, showing that learned value structure did not translate into accuracy gains

Beijing Ingenic Semiconductor Co., Ltd. Jun 2025 – Aug 2025
Machine Learning Engineer Intern

- Developed and optimized machine learning models for embedded and on-chip AI scenarios
- Participated in model training, evaluation, and inference pipeline development using PyTorch
- Assisted in deploying ML models to edge devices with constraints on latency and memory
- Collaborated with algorithm and hardware teams to align model design with chip-level constraints

Benchmark Architectural Design Co., Ltd Jan 2025
Developed MFC front-end modules (UI, event handling, debugging) for a mini-program project in a team setting.

SELECTED PROJECTS

Facere – AI-Native Hardware Design Agent [Website] *with Yiping Zhao, Jiahao Zhang, Zichen Lin*

- Terminal-native agent that drafts and edits schematics and PCB layouts alongside KiCad 9; built a hardware-aware MCP server backed by a curated 153-motif schematic corpus, paired with a PCB physical-simulation package for end-to-end plan–edit–verify loops

TECHNICAL SKILLS

Model Training: RL fine-tuning (GRPO, diffu-GRPO), diffusion language models, LoRA, 4-bit quantization, mixed-precision and multi-GPU training
Research & Evaluation: LLM/VLM reasoning, benchmark design, controlled ablations, multi-seed experiments, calibration and failure analysis
Agent Systems: Multi-agent environments, tool-use workflows, browser/DOM evaluation, MCP
Frameworks & Tools: PyTorch, Hugging Face Transformers, TRL, PEFT, Accelerate, bitsandbytes, DeepSpeed (ZeRO-2), vLLM, W&B, TensorBoard, OpenAI API, Git, Linux, Bash
Programming Languages: Python, C++, MATLAB
Languages: Chinese (Native), English (Fluent)

COMPETITIONS

- ACM-ICPC East Asia Regional Contest – Finalist / Honorable Mention
- National Mathematical Olympiad (China) – Second Prize, 2023
- National Informatics Competition (China) – Second Prize, 2022